Did the Ads Actually Work?

Pages

- Overview
- Targeting & money
- How to measure

Red marks the one thing to look at; blue carries the data.

Independent analysis of public data (Criteo Uplift v2.1, CC BY-NC-SA 4.0; Iowa Liquor Sales). Not affiliated with or endorsed by Criteo, Google or the State of Iowa.

Did the ads actually work?

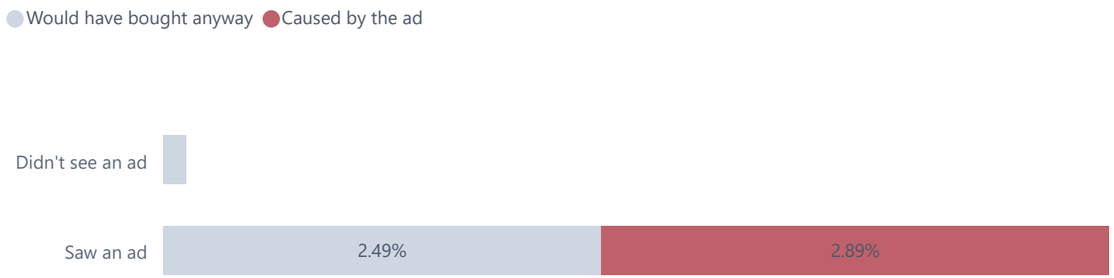
Only 54 of every 100 conversions credited to retargeting were caused by it.

Randomised test · 13,979,592 users · corrected for a small leak

Caused per 100 credited conversions	Seeing an ad: dashboard view	Seeing an ad: real effect	Conversion lift, whole treated group
54	41x	2.2x	+52%
95% interval 50 to 58	Saw an ad vs didn't, as a dashboard shows	From the randomised test	Adjusted (raw +59%)

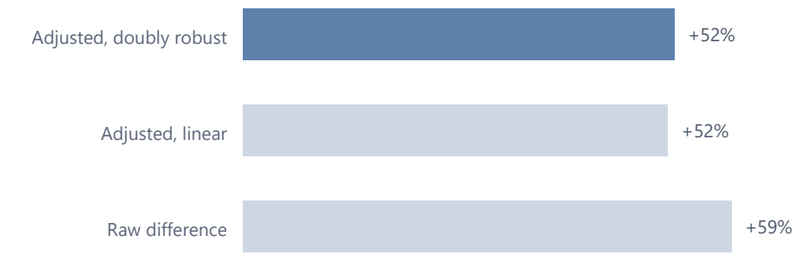
Seeing an ad looks like a 41x boost. The real effect is 2.2x

Conversion rate · users who saw an ad were already 19x likelier to buy; the test separates that from the ad



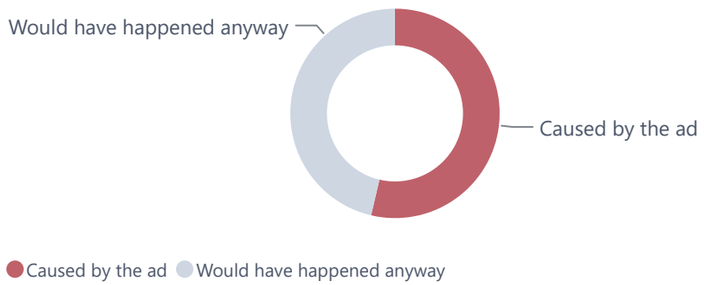
Correcting the leak: conversion lift +59% raw, +52% adjusted

Whole treatment group vs control · features logged before any ad, so adjusting is valid



Of every 100 credited conversions, 54 were caused by the ads

95% interval 50 to 58 · attribution overstates return about 1.9x



How to read this page

Criteo randomly blocked ads for 15% of users. Any difference between the groups was caused by the ads.
Users who saw an ad converted at 5.38%; without it they'd have converted at about 2.49%. So ads work, but far less than a dashboard suggests.
The test had a small leftover imbalance from merging several advertisers' tests; every headline number here is corrected for it (raw figures are shown for comparison).
→ Judge retargeting on what it causes: real ROAS = dashboard ROAS × caused share.

FINDING	EVIDENCE	INTERPRETATION	NEXT ACTION
Only 54 of every 100 conversions credited to retargeting were caused by it.	Seeing an ad lifts conversion from 2.49% to 5.38%: 2.2x, not 41x.	Attribution credits ads with keen shoppers' sales; return is overstated about 1.9x.	Report incremental ROAS next to attributed ROAS in every campaign review.

Source: Criteo Uplift v2.1 (CC BY-NC-SA 4.0) · randomised test, 13,979,592 users · levels describe Criteo's sub-sampled benchmark file

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Who to target, and does it pay?

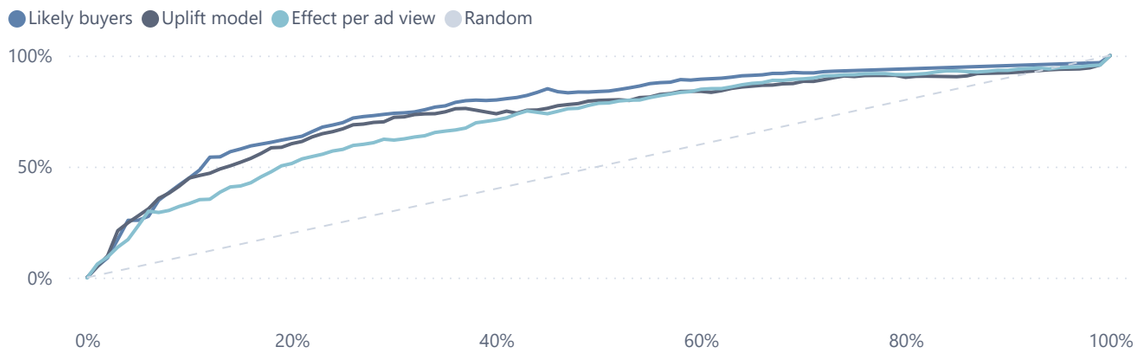
Half the spend, aimed at likely buyers, keeps 84% of the conversions ads cause; the weaker half loses money.

2.8M held-out users · money figures illustrative (ROAS 8, margin 30%)

Effect in the top 20% of users	Half the spend keeps	Real profit per £1 of ads	Weaker half: profit per £1
91%	84%	+£0.29	-£0.58
Of the ads' effect on conversions	Aimed at likely buyers; 95% interval 78% to 90%	Dashboard implies +£1.40 (illustrative)	The half of spend on the least likely buyers

Half the ad spend, aimed at likely buyers, keeps 84% of the conversions ads cause

Share of the ads' effect on conversions captured vs share of spend (ad views) used, best users first



Real profit per £1 of ads, all spend

By dashboard ROAS (rows) and margin (columns) · red = loss · illustrative

Dashboard ROAS	20%	30%	40%	50%
4	-£0.57	-£0.36	-£0.14	+£0.07
6	-£0.36	-£0.03	+£0.29	+£0.61
8	-£0.14	+£0.29	+£0.72	+£1.15
10	+£0.07	+£0.61	+£1.15	+£1.68
12	+£0.29	+£0.93	+£1.58	+£2.22
15	+£0.61	+£1.41	+£2.22	+£3.02

The weaker half of spend loses money in 21 of 24 combinations

Real profit per £1 on the half of spend aimed at the least likely buyers · red = loss

Dashboard ROAS	20%	30%	40%	50%
4	-£0.86	-£0.79	-£0.72	-£0.65
6	-£0.79	-£0.69	-£0.58	-£0.48
8	-£0.72	-£0.58	-£0.44	-£0.30
10	-£0.65	-£0.48	-£0.30	-£0.13
12	-£0.58	-£0.37	-£0.17	+£0.04
15	-£0.48	-£0.22	+£0.04	+£0.30

Break-even dashboard ROAS: 6.2 at a 30% margin

The weaker half needs 19.2 · no ROAS assumption needed

Margin	all spend	best half	other half
20%	9.3	5.6	28.7
30%	6.2	3.7	19.2
40%	4.7	2.8	14.4
50%	3.7	2.2	11.5

FINDING
A fifth of users hold 91% of the effect; half the spend keeps 84%.

EVIDENCE
Uplift models graded on 2.8M held-out users by spend; a simple likely-buyer model matched them.

INTERPRETATION
The weaker half of spend makes -£0.58 per £1 on illustrative numbers.

NEXT ACTION
Test cutting the weaker half with its own holdout before cutting it.

Source: Criteo Uplift v2.1 (CC BY-NC-SA 4.0) · randomised test, 13,979,592 users · levels describe Criteo's sub-sampled benchmark file · money uses stated assumptions · spend measured as ad views

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How to measure it next time

User-level holdouts see small effects; geo tests on noisy sales only catch big ones.

2,700 simulated campaigns on real Iowa sales

Users for a +10% conversion lift
4.9M
1.8 million with a 50/50 split

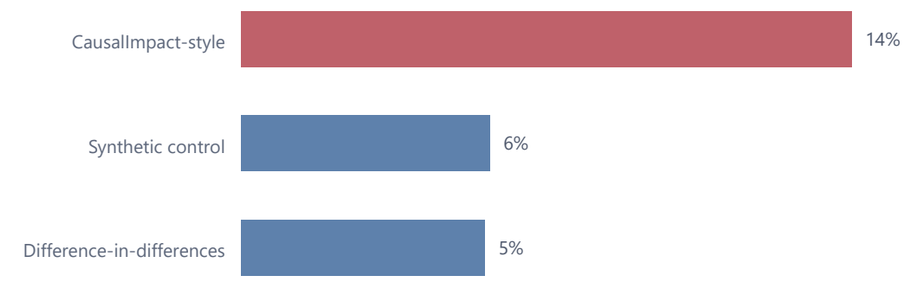
Smallest lift a geo test catches
9%
Best design: 20 counties for 12 weeks

Difference-in-differences false alarms
5%
3% to 8% across designs; promised 5%

CausalImpact-style false alarms
14%
10% to 18%: cries wolf

The CausalImpact-style model cries wolf 2 to 3.5 times too often

Share of no-campaign tests flagged as significant, average of 9 designs · promised: 5%



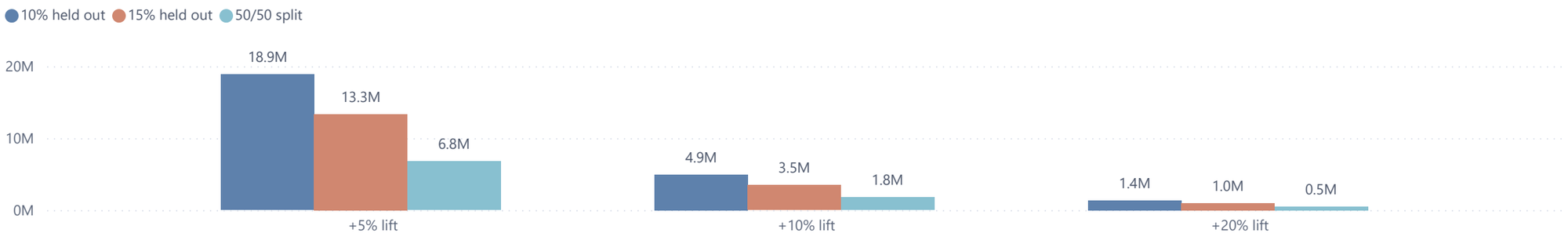
Even 20 counties for 12 weeks only reliably detect about 9%

Difference-in-differences: smallest lift detected 80% of the time · real Iowa sales, simulated campaigns

Test design	4 weeks	8 weeks	12 weeks
5 counties	23%	19%	16%
10 counties	20%	13%	13%
20 counties	12%	10%	9%

Rare conversions need big holdouts: about 4.9 million users to see a +10% lift with a 10% holdout

Users needed to detect a conversion lift 80% of the time (5% false-alarm rate) · 1.8 million with a 50/50 split · base rate from the Criteo control group



FINDING
Holdouts see small effects; geo tests on noisy weekly sales only catch big ones.

EVIDENCE
Best geo design detects about 9%; CausalImpact-style raised 10% to 18% false alarms.

INTERPRETATION
Rare conversions need scale: 4.9 million users for a +10% lift.

NEXT ACTION
Keep an always-on holdout; geo tests only where holdouts are impossible, sized by simulation.

Sources: Iowa Liquor Sales (BigQuery public data), real sales with simulated campaigns · holdout sizing from Criteo control-group rates